

## CHEORWON, Korea, Republic of

WMO: 470950

Lat: 38.1478N

Lon: 127.3042E

Elev: 509

StdP: 14.43

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|------------|------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |            |            | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>0.9 | (c)<br>4.9 | (d)<br>-11.7                  | (e)<br>3.0 | (f)<br>11.9 | (g)<br>-7.9 | (h)<br>3.7 | (i)<br>12.4 | (j)<br>15.7           | (k)<br>28.3 | (l)<br>13.8 | (m)<br>29.6 | (n)<br>1.7            | (o)<br>20 | (p)<br>0.416 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>13.9            | (c)<br>88.6     | (d)<br>75.1 | (e)<br>86.1 | (f)<br>73.5 | (g)<br>83.8 | (h)<br>72.2 | (i)<br>78.2         | (j)<br>84.3 | (k)<br>76.9 | (l)<br>82.5 | (m)<br>75.6 | (n)<br>80.8 | (o)<br>5.3           | (p)<br>230 |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 76.5                            | 140.9 | 81.2 | 75.2 | 134.8 | 80.1 | 74.0 | 129.3 | 79.1 | 42.2          | 84.6 | 40.8 | 82.5 | 39.6 | 81.1 | 83.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 15.4              | 13.2 | 11.6 | -4.5                       | 91.9 | 5.8                | 1.9 | -8.6                                               | 93.3 | -12.0      | 94.4 | -15.2      | 95.4 | -19.4      | 96.8 |
|                   |      |      | -5.2                       | 79.9 | 5.6                | 1.5 | -9.3                                               | 81.0 | -12.6      | 81.8 | -15.7      | 82.7 | -19.8      | 83.8 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 51.1   | 22.0  | 28.0  | 38.7  | 50.9  | 61.8  | 70.2  | 75.4  | 75.9  | 66.7  | 54.5  | 40.5  | 26.7 |
|                                                                                  | DBStd       | 19.72  | 8.26  | 7.63  | 6.60  | 6.73  | 4.82  | 3.88  | 3.97  | 4.35  | 4.99  | 6.73  | 7.95  | 8.18 |
|                                                                                  | HDD50       | 2962   | 868   | 616   | 355   | 69    | 0     | 0     | 0     | 0     | 0     | 34    | 299   | 721  |
|                                                                                  | HDD65       | 6026   | 1333  | 1036  | 815   | 425   | 122   | 7     | 0     | 1     | 38    | 327   | 736   | 1186 |
|                                                                                  | CDD50       | 3348   | 0     | 0     | 4     | 95    | 366   | 606   | 786   | 803   | 500   | 175   | 13    | 0    |
|                                                                                  | CDD65       | 937    | 0     | 0     | 0     | 1     | 22    | 163   | 321   | 339   | 88    | 3     | 0     | 0    |
|                                                                                  | CDH74       | 6630   | 0     | 0     | 0     | 33    | 283   | 1095  | 2167  | 2532  | 502   | 18    | 0     | 0    |
|                                                                                  | CDH80       | 1646   | 0     | 0     | 0     | 1     | 33    | 235   | 563   | 755   | 59    | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.9    | 3.0   | 3.6   | 4.8   | 5.4   | 4.9   | 4.3   | 4.3   | 3.9   | 3.6   | 3.1   | 3.3   | 2.8  |
| Precipitation                                                                    | PrecAvg     | 55.4   | 0.7   | 1.1   | 1.3   | 2.8   | 4.0   | 5.4   | 17.0  | 12.8  | 5.4   | 2.0   | 1.9   | 1.0  |
|                                                                                  | PrecMax     | 87.2   | 2.1   | 3.3   | 3.5   | 6.2   | 11.5  | 15.7  | 39.2  | 32.6  | 17.7  | 6.9   | 4.8   | 2.9  |
|                                                                                  | PrecMin     | 27.1   | 0.0   | 0.0   | 0.2   | 0.5   | 0.5   | 1.6   | 5.7   | 2.2   | 0.5   | 0.3   | 0.4   | 0.1  |
|                                                                                  | PrecStd     | 12.2   | 0.5   | 0.8   | 0.9   | 1.4   | 2.1   | 3.4   | 8.3   | 8.1   | 4.4   | 1.5   | 1.0   | 0.7  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 44.6  | 53.8  | 65.1  | 78.3  | 83.7  | 88.4  | 91.4  | 91.4  | 84.6  | 76.3  | 64.7  | 49.5 |
|                                                                                  |             | MCWB   | 38.5  | 43.2  | 49.4  | 56.8  | 61.1  | 67.6  | 77.5  | 77.1  | 70.6  | 61.4  | 54.5  | 43.1 |
|                                                                                  | 2%          | DB     | 40.5  | 48.2  | 59.9  | 73.3  | 80.1  | 85.2  | 88.1  | 89.2  | 81.6  | 72.6  | 60.3  | 45.6 |
|                                                                                  |             | MCWB   | 34.9  | 38.6  | 46.7  | 54.9  | 60.3  | 67.2  | 75.4  | 76.6  | 68.7  | 59.7  | 51.4  | 40.3 |
|                                                                                  | 5%          | DB     | 37.7  | 44.1  | 55.8  | 69.1  | 77.3  | 82.7  | 85.5  | 86.9  | 79.3  | 69.5  | 57.2  | 42.3 |
|                                                                                  |             | MCWB   | 32.6  | 36.5  | 43.6  | 52.6  | 59.5  | 66.7  | 74.3  | 75.7  | 67.1  | 57.9  | 49.9  | 37.4 |
|                                                                                  | 10%         | DB     | 35.0  | 40.5  | 51.9  | 65.5  | 74.2  | 80.3  | 83.1  | 84.3  | 77.0  | 66.6  | 53.9  | 39.2 |
|                                                                                  |             | MCWB   | 30.7  | 33.8  | 41.8  | 51.0  | 58.3  | 65.9  | 73.6  | 74.5  | 65.6  | 56.1  | 47.6  | 34.8 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 39.7  | 45.7  | 53.4  | 62.7  | 66.5  | 73.3  | 79.8  | 80.6  | 75.2  | 65.5  | 57.2  | 45.6 |
|                                                                                  |             | MCDB   | 43.2  | 49.3  | 59.8  | 70.4  | 74.5  | 80.3  | 86.3  | 87.7  | 80.6  | 71.0  | 61.5  | 48.1 |
|                                                                                  | 2%          | WB     | 36.1  | 40.1  | 48.9  | 58.1  | 64.3  | 71.4  | 78.0  | 78.7  | 72.6  | 62.3  | 54.0  | 40.8 |
|                                                                                  |             | MCDB   | 39.5  | 46.8  | 56.4  | 67.0  | 73.0  | 78.1  | 84.1  | 84.8  | 77.4  | 68.6  | 58.3  | 44.3 |
|                                                                                  | 5%          | WB     | 33.4  | 37.2  | 45.5  | 55.3  | 62.5  | 70.0  | 76.6  | 77.4  | 70.4  | 60.3  | 51.1  | 38.0 |
|                                                                                  |             | MCDB   | 36.5  | 42.8  | 53.3  | 64.6  | 70.9  | 76.7  | 81.9  | 83.4  | 75.7  | 66.2  | 55.7  | 41.7 |
|                                                                                  | 10%         | WB     | 31.4  | 34.7  | 42.6  | 52.8  | 60.9  | 68.7  | 75.5  | 76.3  | 68.3  | 58.2  | 48.1  | 35.2 |
|                                                                                  |             | MCDB   | 34.5  | 39.5  | 50.7  | 62.6  | 69.6  | 75.6  | 80.5  | 81.9  | 73.9  | 64.5  | 52.5  | 38.6 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 20.2  | 20.6  | 20.7  | 23.1  | 20.8  | 17.4  | 12.3  | 13.9  | 18.5  | 22.1  | 19.0  | 18.3 |
|                                                                                  |             | MCDBR  | 20.1  | 24.1  | 27.4  | 29.4  | 27.0  | 22.6  | 16.9  | 17.5  | 21.1  | 24.1  | 22.9  | 19.7 |
|                                                                                  |             | MCWBR  | 14.5  | 15.3  | 15.0  | 13.2  | 10.0  | 7.8   | 6.6   | 6.8   | 9.1   | 11.4  | 14.2  | 14.2 |
|                                                                                  | 5% WB       | MCDBR  | 16.5  | 20.6  | 23.0  | 23.0  | 19.5  | 15.6  | 12.5  | 14.3  | 15.2  | 19.6  | 18.3  | 18.1 |
|                                                                                  |             | MCWBR  | 13.1  | 14.3  | 14.6  | 12.4  | 9.4   | 6.9   | 5.8   | 6.4   | 8.0   | 10.9  | 13.3  | 14.0 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.407  | 0.469 | 0.553 | 0.607 | 0.613 | 0.680 | 0.653 | 0.561 | 0.489 | 0.485 | 0.460 | 0.407 |      |
|                                                                                  | taud        | 1.947  | 1.841 | 1.724 | 1.659 | 1.687 | 1.614 | 1.726 | 1.948 | 2.067 | 1.950 | 1.947 | 1.990 |      |
|                                                                                  | Ebn at Noon | 228    | 230   | 224   | 222   | 224   | 209   | 213   | 231   | 240   | 223   | 209   | 218   |      |
|                                                                                  | Edh at Noon | 43     | 55    | 68    | 77    | 76    | 82    | 73    | 57    | 48    | 49    | 43    | 39    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 776    | 1036  | 1322  | 1552  | 1742  | 1626  | 1250  | 1323  | 1284  | 1112  | 758   | 690   |      |
|                                                                                  | RadStd      | 56     | 73    | 132   | 126   | 167   | 153   | 185   | 178   | 145   | 80    | 82    | 57    |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +176  | N/A   |
| (47) Regional (41 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +189  | +103  |

Nomenclature: See separate page